

NDONDA DANIEL MATONDO

Software Engineer | Full-Stack & Backend Developer

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PROFESSIONAL SUMMARY

Software Engineer with a solid foundation in Systems Engineering, Backend Development, and Observability. Completed the **42 Luanda Common Core** with deep proficiency in Unix environments and low-level C/C++ programming, alongside practical experience in modern stacks (Python/FastAPI, Node.js, React, Docker) and Computer Vision. Proven track record in technical leadership at hackathons and delivering resilient, real-time distributed systems.

EDUCATION

Software Engineering (Common Core Completed) 2024 – Present | Luanda, Angola
42 Luanda
Rigorous project-based curriculum focused on Data Structures & Algorithms (DSA), low-level C/C++ programming, concurrency, and distributed system architectures without external third-party libraries.

Technical High School Diploma in Computer Science 2020 – 2024 | Luanda, Angola
Instituto Médio Industrial Simione Mucune (IMISM)
Foundations of computer algorithms, programming logic, operating systems, computer networking, and relational database modeling.

EXPERIENCE & LEADERSHIP

Tech Lead & Backend Architect — InfraWatch 2025
RCS & 42 Luanda Hackathon (Finalist)

- Led the technical architecture and engineering standards, defining API contracts and coordinating milestone delivery under strict deadlines.
- Engineered real-time backend microservices in FastAPI (Python) for metric collection, stream processing, and infrastructure monitoring.
- Orchestrated containerized environments using Docker, ensuring isolated dependency management and rapid deployment pipelines.

Freelance Software Engineer 2024 – 2025
Miss Xtrela & Commercial Solutions

- Miss Xtrela (Inventory & Biometrics System): Developed a complete desktop application (PySide6/SQLite) with facial biometric authentication via OpenCV (LBPH / Haar Cascades) and sales analytics, automating enterprise inventory tracking.
- Fuel Management System: Designed secure RESTful APIs with JWT authentication and RBAC permissions, including MySQL database schema modeling for gas station operations.

FEATURED ENGINEERING PROJECTS

BidLive — Real-Time Live Auction Platform (ft_transcendence) 2025
React, TypeScript, FastAPI, Node.js, Docker, WebSockets, WebRTC (LiveKit), PostgreSQL, Redis, Prometheus, Grafana, ELK

- Engineered full microservices infrastructure and containerization with Docker Compose and NGINX reverse proxy (TLS/SSL), integrating PostgreSQL and Redis.
- Implemented end-to-end Observability stack using Prometheus, Grafana, Alertmanager, and ELK Stack (Elasticsearch, Logstash, Kibana) for system telemetry and metrics.
- Architected real-time bidirectional communication and messaging with WebSockets and Redis Pub/Sub for live bidding broadcasts and interactive chat.

Webserv — High-Performance HTTP/1.1 Web Server 2024 – 2025
C++98, epoll/poll, Unix Network Sockets, CGI, HTTP/1.1 RFC

- Built a non-blocking HTTP/1.1 web server from scratch utilizing I/O multiplexing (epoll/poll), handling concurrent requests with high throughput and zero memory leaks.
- Implemented full support for REST methods (GET, POST, DELETE), multipart file uploads, and dynamic CGI script execution.

Minishell & Inception (Systems Engineering & DevOps) 2024
C Language, Docker Compose, Linux/Unix, POSIX, Bash, NGINX TLS, MariaDB

- Minishell (C): Engineered a Unix command interpreter with process management (fork, execve), concurrent pipelines, I/O redirection, and asynchronous POSIX signal handling.
- Inception: Orchestrated a multi-container service infrastructure on Alpine Linux using Docker Compose, isolated networks, and persistent data volumes.

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript (ES6+), TypeScript, SQL, HTML5/CSS3.

Backend & APIs: FastAPI, Node.js, RESTful APIs, WebSockets, Redis (Pub/Sub & Cache), JWT, OAuth 2.0 / OIDC, RBAC, CGI.

Frontend & Desktop: React, Tailwind CSS, Vanilla JavaScript, Responsive UI & Component Architecture, PySide6 / PyQt.

Data, AI & Computer Vision: OpenCV (LBPH / Haar Cascades), Pandas, NumPy, Matplotlib, Genetic Algorithms (NEAT), Web Scraping (BeautifulSoup, Selenium).

Databases: PostgreSQL, MySQL, MongoDB, Redis, SQLite, Relational Modeling & Query Optimization.

DevOps & Observability: Docker, Docker Compose, Linux/Unix, NGINX, GitHub Actions (CI/CD), Prometheus, Grafana, ELK Stack.

LANGUAGES & HONORS

Languages: Portuguese (Native) | English (Intermediate B1 - Technical Reading & Professional Writing).

Honors & Awards: Finalist – RCS & 42 Luanda Hackathon (Project InfraWatch).